

Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

Jagadeesh Rachapudi, Ramakrishna Vamsi Setti, Ritoli Vatsi, Praful Hambarde, Sachin Chaudhary,
and Amit Shukla

Abstract—Machine unlearning for privacy-critical biometric systems, fingerprint, iris, and face recognition, remains underexplored despite GDPR/CCPA mandates requiring erasure throughout deployment. Existing methods target classification and generative tasks, collapsing all forget classes to a single distant point, which inflates convergence time on resource-constrained edge devices facing frequent, minimal-replay removal requests, and they are also benchmarked almost exclusively on ResNet/ViT for CIFAR/ImageNet, leaving biometric backbones untested. We propose Camouflage Unlearning (CU), which maps forget classes to nearby retain-class boundaries instead of a single point, reducing migration distance. This proximity target shapes only the training-time optimization objective, and we verify empirically that it does not induce elevated impostor matches against the assigned retain identity at verification time. CU achieves 3–4× faster convergence via retain-dominance constraints and localized regularization that surgically updates specific transformer blocks, while maintaining competitive retain accuracy. Across classification, iris, fingerprint, and face recognition, CU shows competitive performance and privacy behavior, with MIA ≈ 0.5 , supporting its effectiveness for regulatory-compliant, deep embedding-based biometric authentication.

Index Terms—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers

I. INTRODUCTION

MACHINE unlearning (MU) has emerged as a critical research area [1], [2], addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [3]. The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [4], [5], [6]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [7], [8] mandate that model service providers erase user data from learned representations upon request, establishing legal and ethical obligations for responsible AI deployment. Users possess the right to request data removal at any time, so per-request efficiency becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [9] and generative tasks [10], with limited exploration of privacy-critical biometric systems. While

face recognition [11], [12], [13] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [14] or gradient ascent [15]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [16], geometric boundary manipulation [17], perturbation-based forgetting [18], and meta-learning optimization [19], [20]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Methods such as GS-LoRA [15], SCRUB [14], SalUn [16], Boundary Unlearning [17], Forget Vectors [18], MCU [20], LTU [19], DELETE [21], LoTUS [22], and MUNBa [23] each trade off convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [24] or ViT backbones [25] for CIFAR [26] and ImageNet [27]. Their transferability to biometric backbones remains untested, despite architectures such as Face Transformer [28], LAM [29], JIPNet [30], SSLDMNet-Iris [31], IrisFormer [32], PFSimNet [33], and MEM-FR [34] representing the state of the art among deep embedding-based biometric backbones. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4× faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. This proximity target governs only the training-time optimization objective, not final

Jagadeesh Rachapudi, Ritoli Vatsi, Praful Hambarde, and Amit Shukla are with the Drone Lab, Indian Institute of Technology Mandi, Himachal Pradesh, India. E-mail: s23096@students.iitmandi.ac.in; ritoli245994@gmail.com; praful@iitmandi.ac.in; amitshukla@iitmandi.ac.in

Ramakrishna Vamsi Setti is with the Indian Institute of Science (IISc), Bengaluru, India. E-mail: vamsiseti007@gmail.com

Sachin Chaudhary is with the Centre of Excellence for Artificial Intelligence, School of Computer Science, UPES, Dehradun 248007, India. E-mail: sachin.chaudhary@ddn.upes.ac.in

verification-level similarity. A naive reading might suggest CU induces impersonation of the nearest retain identity, but Section III-D empirically verifies that forget probes do not match their assigned retain centroid more often than chance, ruling out this concern. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network. Additional details are provided in Section II.

Our main contributions are summarized as follows:

- 1) Among the earliest studies of machine unlearning for privacy-critical deep embedding-based biometric systems, covering fingerprint, iris, and face recognition.
- 2) Camouflage Unlearning, achieving 3–4× faster convergence than SoTA methods with competitive retention accuracy and privacy guarantees.
- 3) Empirical evaluation across biometric and classification domains showing CU concentrates updates in specific network blocks and remains effective under data-scarce replay.

II. PROBLEM SETTING AND METHODOLOGY

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), mapping $f_{\mathcal{M}} : \mathcal{X}_D \rightarrow \mathcal{Y}_D$. Since storing D_r^{full} is prohibited under GDPR/CCPA and full retraining is costly, we retain only a small replay buffer $D_r \subset D_r^{full}$, at most 10% of $|D_r^{full}|$ ¹. Before unlearning, $\mathcal{M}$ correctly maps D_f and D_r . Applying $\mathcal{F}$, described in Section II-B, yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists, as shown below.

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \rightarrow \mathcal{Y}_{D_f}; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\rightarrow \mathcal{Y}_{D_f}; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \quad (1)$$

where $\not\rightarrow$ denote forgotten/retained mappings. Successful unlearning requires:

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping, a many-to-many distribution replacing the many-to-one collapse of existing methods, reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|} \sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \dots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \dots, c_{f_M}\}$. Each forget centroid is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg \min_j \|c_{f_i} - c_{r_j}\|_2 \quad (2)$$

We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x}, y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \quad (3)$$

¹From this point, D_r denotes the replay buffer unless specified.

This risks corrupting retain decision regions and prolonging convergence, so we enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU \left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j \right) \quad (4)$$

where z_i are logits and F, R index forget/retain classes. This accelerates forget suppression while preserving retain accuracy. Retain knowledge is further preserved via classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x}, y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \quad (5)$$

where $\mathbf{z}_r$, y_r , and c_y are retain logits, labels, and original centroids. Since full-parameter updates risk instability under limited replay, inspired by group sparse selection [15], we apply localized regularization over fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^G \|\theta_g\|_2 \quad (6)$$

where θ_g are block- g parameters and G is the number of groups, each initialized at zero to track the *change* from pretrained weights, so $\mathcal{L}_{local}$ penalizes update magnitude rather than absolute weights. This group-Lasso approach concentrates updates in few blocks, preserving retained knowledge under constrained replay. The total loss:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom} \mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \quad (7)$$

III. PERFORMANCE EVALUATION

A. Experimental Setup

Datasets, Backbones, and Baselines: We evaluate on CIFAR-100 [26] (DeiT-Tiny [35]), CASIA-Iris [36] (LAM [29]), Sokoto Coventry fingerprints [37] (JIPNet [30]), and CASIA-Face100 [38] (Face Transformer [28]), comparing CU against SCRUB [14], SalUn [16], GS-LoRA [15], LTU [19], MCU [20], and Forget Vectors [18], plus an Oracle trained only on D_r^{full} as the upper bound.

Evaluation Metrics: We report Acc_f , Acc_r , MIA using the label-only attack [39] where ≈ 0.5 is safe, KL divergence to Oracle, RTE, and H-Mean [40]. Success requires $Acc_f \approx 0$, $Acc_r \approx Oracle$, low KL/RTE. $FAR@\{0.3, 0.4, 0.5\}/EER$ via cosine similarity further verify open-set identity confusion in Section III-D.

B. Main Results

Table I compares all methods across four tasks, where **green/blue** mark best/second-best.

Results: All methods forget near-perfectly ($Acc_f < 1\%$). CU ranks first or second in Acc_r and H-Mean across all tasks, with LTU and MCU closest, achieves MIA ≈ 0.5 despite GS-LoRA's lower KL from deliberately reshaped forget logits, and delivers a 3–4× RTE speedup over every baseline.

Geometric Analysis: Figure 1 visualizes embedding-space t-SNE. GS-LoRA [15], SCRUB [14], and Boundary Unlearning [17] collapse forget classes to one distant point. CU instead migrates them to the nearest retain boundary, explaining its RTE gains in Table I.

Method	Classification						Iris recognition					
	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [15]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [16]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [14]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [18]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [20]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [19]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [15]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [16]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [14]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [18]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [20]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [19]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I
PERFORMANCE ACROSS FOUR DOMAINS. CU ACHIEVES 3–4 $\times$ RTE SPEEDUP.

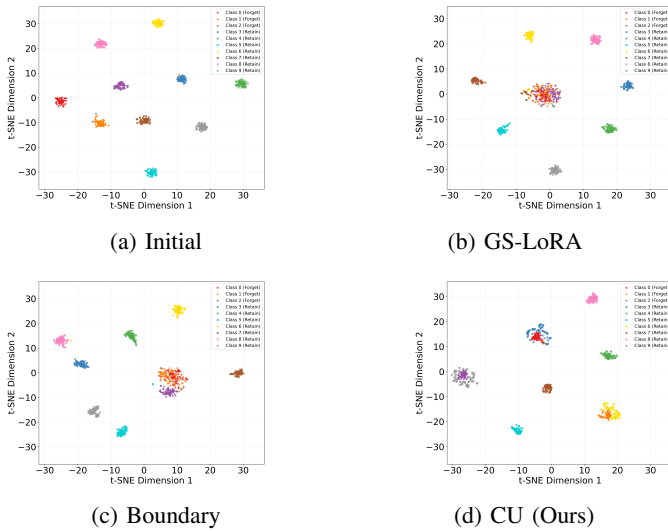


Fig. 1. t-SNE comparison showing CU spreads embeddings while baselines collapse to one point.

C. Ablation Studies

Replay Buffer Size: Table II shows a 0.1 replay buffer ratio optimally balances retention accuracy and efficiency on CIFAR-100.

Buffer Ratio	Speed	Acc_f	Acc_r	$H\text{-Mean}$
1.0	1 $\times$	1.33	74.39	2.62
0.5	2 $\times$	1.18	73.00	2.32
0.3	3.3 $\times$	1.33	74.55	2.61
0.1	10 $\times$	1.30	74.52	2.56

TABLE II
REPLAY BUFFER SIZE ABLATION. THE 0.1 RATIO BALANCES RETENTION AND EFFICIENCY.

Retain Dominance Constraint: Table III shows the constraint reduces Acc_f from 9.97% to 0.39% by eliminating a long-tail convergence delay, with minimal retention loss.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE III
DOMINANCE CONSTRAINT ELIMINATES LONG-TAIL CONVERGENCE.

Localized Parameter Regularization: On CIFAR-100, Figure 2 and Table IV show this regularization concentrates updates in few blocks, reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE IV
LOCALIZED REGULARIZATION LOWERS ℓ_2 NORM WITH MINIMAL ACCURACY LOSS.

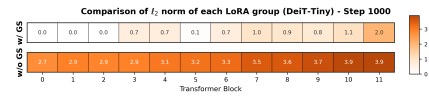


Fig. 2. Per-block ℓ_2 norm, showing updates concentrate in few blocks.

Scalability: Table V shows CU maintains consistent forgetting and retention across five ViT backbones, from DeiT-Tiny to ViT-Large.

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE V
ViT BACKBONE SCALE VS. UNLEARNING PERFORMANCE.

Sequential Removal Workload: We pilot CU against

DELETE [21], LoTUS [22], MUNBa [23], and LTU [19] across successive forget cycles on CIFAR-100 [26] and CASIA-Face100 [38]. Figure 3 shows baselines degrade progressively while CU retains stable accuracy, confirming robustness under continual removal.

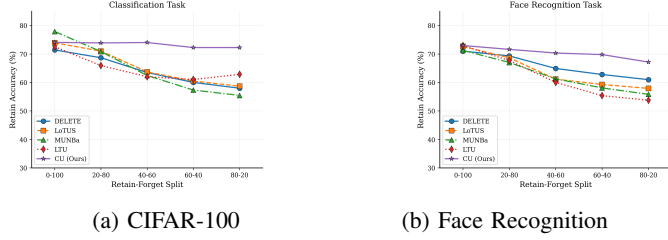


Fig. 3. Sequential-cycle retain accuracy, showing baselines degrade while CU stays stable.

D. Attack-Based Verification

Table VI reports FAR/EER of forget probes vs. the retain gallery on CASIA-Iris. CU tracks the Oracle closely, with FAR@0.5 at 3.83% vs. 3.10% and EER at 21.99% vs. 21.69%, and both rise similarly over *Before* at 1.64%, so the small increase stems from removing identities, not from CU.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE VI

FAR/EER OF FORGET PROBES VS. RETAIN GALLERY ON CASIA-IRIS.

Aggregate FAR/EER cannot rule out a concentrated, backdoor-like confound between one forget identity and its assigned nearest-retain centroid, since such a pairing barely moves the aggregate rate. Table VII reports this per-identity measurement on CASIA-Face100 against the Oracle, with match rates in % at the same cosine-similarity threshold as FAR@0.5 in Table VI, covering assigned-centroid match, linear probe, relearning, MIA, and model inversion. CU's assigned-centroid match rate is no higher than Oracle's, ruling out concentrated confounding between individual forget-retain pairs. Its linear-probe rate stays within a point of Oracle, relearning is slow, and label-only/adaptive MIA plus inversion match sit near chance, showing forgetting is diffuse and adaptive attacks also fail.

Test	CU	Oracle
Assigned-centroid match ↓	2.71%	3.54%
Linear probe (frozen emb.) ↓	5.28%	4.85%
5-shot relearn Acc_f ↓	4.37%	7.12%
MIA, label-only [39] ≈ 0.5	0.52	0.51
MIA, adaptive ≈ 0.5	0.54	0.54
Model inversion match ↓	3.42%	3.78%

TABLE VII

PER-IDENTITY CONFOUND TESTS.

Attention Maps: Figure 4 shows retrained and CU models shift attention off-face *only for forget* CASIA-Face100 [38] samples, confirming surgical identity removal.

Relearning Attacks: Figure 5 shows CU recovers gradually versus a head-masked baseline's rapid recovery under a relearning attack, confirming genuine backbone-level forgetting.

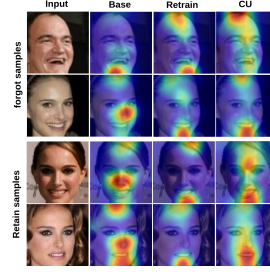


Fig. 4. Attention maps showing retrained/CU models disperse attention off-face only for forget identities.

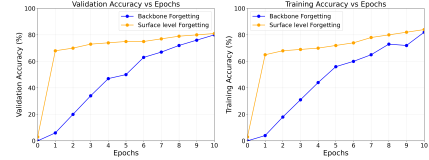


Fig. 5. CU recovers gradually as genuine forgetting, while head-masking recovers rapidly as superficial forgetting.

IV. RESULTS DISCUSSION

A. On the Necessity of the Retain Buffer

A natural concern is whether the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR/CCPA erasure mandates. We argue D_r is unavoidable, because pretraining entangles D_f and D_r^{full} in the same parameters, so disentangling them needs a retained reference unless architectural support such as modular pathways is present. Alternatives fall short. Impair-and-repair [41] still needs D_r for repair, source-free unlearning [42] works only for convex/linear models, and other non-convex methods also rely on D_r [43], [16], [14], [15], [40], [44], [45], [46], [47], with only the generative-task exceptions FLAT [48] and ESD [49] avoiding it. CU therefore keeps a minimal D_r , at most 10% of D_r^{full} with at least one sample per retain identity, computes retain metrics on the full held-out split rather than D_r , and holds only *consenting* identities, so a withdrawal is treated as a forget request and the buffer refills from the consenting pool.

B. Scope Across Modalities

While face recognition is dominated by deep embedding models, several deployed iris and fingerprint systems still rely on classical, non-learned representations, such as Daugman iris codes and minutiae matching. The reported gains of CU apply to deep embedding-based biometric backbones. Extending camouflage unlearning to classical, non-differentiable feature pipelines is outside the current scope.

V. CONCLUSION

We address machine unlearning for privacy-critical deep embedding-based biometric systems, spanning fingerprint, iris, and face recognition, where existing many-to-one collapse causes prolonged convergence under frequent edge-device unlearning requests. Our Camouflage Unlearning achieves up to 4× speedup via many-to-many distribution and surgical parameter modifications, with forgetting, retention, and privacy comparable to state-of-the-art methods. Future work extends CU to continual forgetting in pretrained biometric models.

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